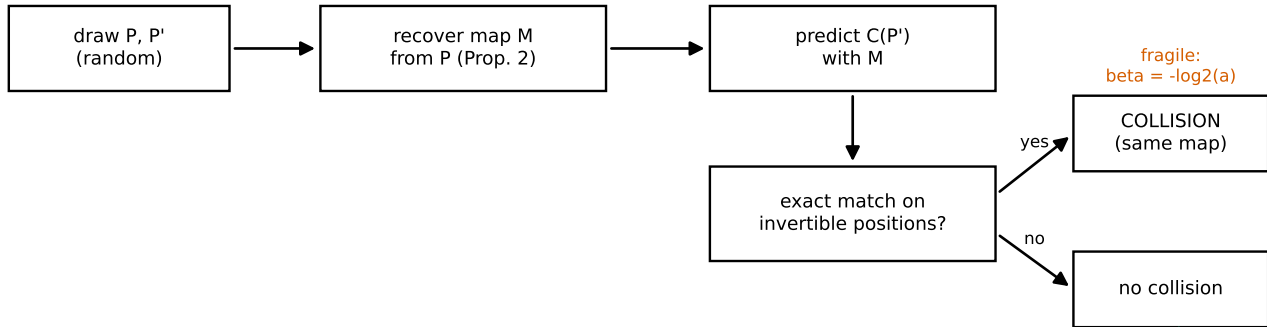




fragile:
 $\beta = -\log_2(a)$

COLLISION
(same map)

exact match on
invertible positions?

yes

no

no collision

genuine:
 $\beta \geq 2 \log_2(k)$

$a = \text{map-collision rate over } k \text{ draws (birthday: } a \sim 1/N_{\text{eff}}, \beta = \log_2 N_{\text{eff}})$